

Where Optimal Networks Stop Existing: Certified Spectral Limits of Cyclic Architectures

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1. Research Question

- Every graph has an **Ihara zeta function**, so one can state the **Riemann Hypothesis** for a graph — and unlike the classical case it is **decidable**: a cubic graph satisfies it exactly when every eigenvalue of modulus $\neq 3$ has modulus $\leq 2\sqrt{2}$.
- **Which generalized Petersen graphs satisfy it?** Their spectrum is known; the question is not asked.

2. Methodology / Project Design

- **Clear the radicals.** Substituting $\pm 2\sqrt{2}$ into the characteristic polynomial and squaring twice (each an *equivalence*) leaves one integer polynomial D_k of degree $2k + 2$, evaluated on the grid.
- **Read the endpoints.** $D_k(1) = -7$ always; $D_k(-1) = -7$ for odd k , $+9$ for even k . Each forbids a band the grid must miss.
- **Certify exactly** in $\mathbb{Z}[\zeta_m]$.

3. Data Analysis & Results

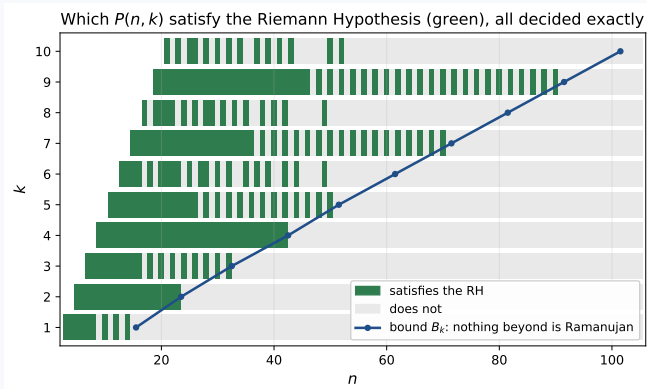


Figure: author's own work, Python and T_EX, 2026. Green: satisfies the RH. Line: the certified bound B_k .

4. Interpretation & Conclusions

- **Only finitely many n per k** , and proved for **every k** : $n \leq \pi(2 + \sqrt{2})\sqrt{k^2 + 1}$, by AM–GM and $1 - \cos \theta \leq \theta^2/2$.
- **Parity law**: for odd k an even n lands on an exempt eigenvalue, so **bipartiteness helps**; odd n stops at half the bound.
- **Criterion with no k in it**: both factors of D_k collapse to $Q = 4x^2 + 4y^2 - 8\sqrt{2}|xy| + 3 \geq 0$ — one fixed forbidden set for every k .
- **Complete classification**, every n and every k : 460 pairs (324 up to isomorphism)